Lines

SAT Math · Foundations to Mastery · Week of 4 October

Name: _____ Due: _____

How to use this sheet

This is four short sittings, not one long one. Four half-hours across the week will do more for you than one long Saturday, and you are far more likely to finish it.

- **Part A is no calculator.** None. It is 15 minutes and it keeps last week's sign work alive.
- **Part D is the part that pays.** It asks what a number *means*. Read each stem twice before you answer.
- When you get one wrong, write it in your error log: what you did, and what you should have done.

Sitting	Part	What it drills	Minutes
Sunday	A	Sign retention, no calculator	15
Tuesday	B	Solving linear equations	25
Thursday	C	Slope, four ways	25
Saturday	D, E	What the number means; six mixed, timed	25

Part A — Sign retention

No calculator. 15 minutes.

SUNDAY

Last week you fixed this. Keep it fixed. **Write every substitution in brackets** before you work anything out — that single habit is what stopped the bleeding.

1. $8-(-5)=$ _____
2. $(-7)^2=$ _____
3. $-7^2=$ _____
4. $-6-(-6)=$ _____
5. $\dfrac{-24}{-6}-10=$ _____
6. Expand: $-4(x-3)$ _____
7. Expand and simplify: $-(5x-9)+2x$ _____
8. $f(x)=x^2-3x$. Find $f(-4)$. _____
9. $g(x)=5-2x^2$. Find $g(-3)$. _____
10. Solve $4-3x>13$. _____

Part B — Solving linear equations

Calculator allowed. 25 minutes.

TUESDAY

One move per line, and write what you did in the margin. When the variable disappears from both sides, stop and read what is left: a false statement means no solution, a true statement means every number works.

1. $5(x-3) = 2x+9$. What is the value of x?
   Answer: _____ *(write your answer)*

2. $7x-4 = 3x+20$. What is the value of x?
   Answer: _____ *(write your answer)*

3. $2(3x+1) = 4(x-5)$. What is the value of x?
   Answer: _____ *(write your answer)*

4. $-2(x-4) = 3x+13$. What is the value of x?
   Answer: _____ *(write your answer)*

5. $\frac{x}{3}+\frac{x}{4} = 14$. What is the value of x?
   Answer: _____ *(write your answer)*

6. $\frac{2x-1}{5} = \frac{x+4}{3}$. What is the value of x?
   Answer: _____ *(write your answer)*

7. $\frac{3}{4}x-2 = \frac{1}{2}x+1$. What is the value of x?
   Answer: _____ *(write your answer)*

8. The perimeter of a rectangle is $P = 2\ell + 2w$, where ℓ is the length and w is the width. Which equation gives w in terms of P and ℓ?
   A) $w = P - 2\ell$ B) $w = \frac{P-2\ell}{2}$ C) $w = \frac{P}{2} - 2\ell$ D) $w = \frac{P-\ell}{2}$

9. How many solutions does $6x+5 = 2(3x+1)+3$ have?
   A) Zero B) Exactly one C) Exactly two D) Infinitely many

10. How many solutions does $4(2x-3) = 8x+7$ have?
    A) Zero B) Exactly one C) Exactly two D) Infinitely many

Part C — Slope, four ways

Calculator allowed. 25 minutes.

THURSDAY

Slope is one idea wearing four costumes: two points, a table, a graph or equation, and a line you have to write yourself. Always subtract in the same order top and bottom.

1. A line passes through $(-3,-5)$ and $(1,7)$. What is its slope?

   Answer: _____ *(write your answer)*

2. A line passes through $(2,-1)$ and $(-4,8)$. What is its slope?

   Answer: _____ *(write your answer)*

3. The table shows four points on a line.

x	-2	1	4	7
y	11	2	-7	-16

   What is the slope of the line?

   A) -9 B) -3 C) $-\frac{1}{3}$ D) 3

4. The table shows four points on a line.

x	0	3	6	9
y	-4	2	8	14

   Which equation describes the line?

   A) $y = 2x - 4$ B) $y = 2x + 4$ C) $y = 6x - 4$ D) $y = -4x + 2$

5. In the xy-plane, a line passes through $(0,7)$ and falls 2 units for every 5 units it moves to the right. What is the slope of the line?

   Answer: _____ *(write your answer)*

6. A line is given by $4x + 3y = 9$. What is its slope?

   A) 4 B) $-\frac{3}{4}$ C) $\frac{4}{3}$ D) $-\frac{4}{3}$

7. A line passes through $(-2,5)$ and has slope -3. Which equation describes it?

   A) $y = -3x + 11$ B) $y = -3x - 1$ C) $y = -3x + 5$ D) $y = -2x + 5$

8. A line is given by $3x - 2y = 12$. What is the y-coordinate of its y-intercept?

   Answer: _____ *(write your answer)*

Part D — What the number means

Calculator only where asked. 15 minutes.

SATURDAY

Answer these in words, or by choosing a statement. **Do not calculate** unless an item tells you to. In every model below, one number is the starting amount and one number is the rate of change. Decide which is which before you write anything.

This is the highest-value part of the sheet. On the diagnostic you did the arithmetic correctly and still lost the mark, because the question was asking something else.

1. A repair company charges $C = 0.18m + 45$ dollars for a visit, where m is the number of miles driven to reach the customer. What does the 45 represent?

   _____

2. For the same model $C = 0.18m + 45$, what does the 0.18 represent?

   _____

3. A savings balance is modelled by $P = 250 + 12w$ dollars, where w is the number of weeks since the account was opened. Which statement is correct?

   A) The balance grows by \$250 each week. B) The balance grows by \$12 each week.

   C) The balance reaches \$12 after 250 weeks. D) The account was opened with \$12.

4. The value of a delivery van is modelled by $V = 1800 - 150t$ dollars, where t is the number of years since it was bought. Which statement is correct?

   A) The van loses \$150 of value each year. B) The van gains \$150 of value each year.

   C) The van was worth \$150 when bought. D) The van loses \$1800 of value each year.

5. A gym membership costs $T = 35m + 60$ dollars in total, where m is the number of months. Work out the total cost of an 8-month membership, then say which part of that total does not depend on m.

   Total: _____ Part that does not depend on m: _____

6. A phone plan costs $C = 0.05t + 19$ dollars a month, where t is the number of text messages sent. A student says that doubling the number of texts doubles the monthly cost. In one sentence, say why that is wrong.

   _____

Part E — Mixed, timed

Calculator allowed. 12 minutes, timed.

SATURDAY

Set a timer for 12 minutes. Answer every one — there is no penalty for a wrong answer, so a blank is a mark thrown away. Then check each answer by substituting it back.

1. What value of x satisfies $4(x+3) = 2x+22$?
   A) 5 B) 9.5 C) 17 D) 20

2. A line passes through $(-5, 4)$ and $(3, -2)$. What is its slope?
   Answer: _____ *(write your answer)*

3. A line is given by $2x - 5y = 20$. What is the y-coordinate of its y-intercept?
   A) -4 B) $\frac{2}{5}$ C) 4 D) 10

4. How many solutions does $5(2x-1) = 10x+4$ have?
   A) Zero B) Exactly one C) Exactly two D) Infinitely many

5. A technician's weekly pay is modelled by $W = 4.5h + 28$ dollars, where h is the number of hours worked. What does the 4.5 represent?
   A) The pay in dollars for each hour worked. B) The weekly pay before any hours.
   C) The number of hours worked each week. D) The total weekly pay in dollars.

6. A tank is draining. The volume remaining is modelled by $V = 500 - 12t$ litres, where t is the number of minutes since draining began. How many litres remain after 15 minutes?
   Answer: _____ *(write your answer)*

Error log

Fill this in as you go.

Date	Q	What I did	What I should have done

Bring this sheet and your error log to the next session.